Comment by Massive Bio, Inc. on the Development of an Artificial Intelligence (AI) Action Plan
(Submitted via Regulations.gov on February 10, 2025)

I. Introduction

Massive Bio, Inc. appreciates the opportunity to respond to the National Science Foundation (NSF) Request for Information (RFI) regarding the Development of an Artificial Intelligence (AI) Action Plan ("Plan"). We strongly support the Executive Order of January 23, 2025 (E.O. 14179), which aims to remove barriers to American leadership in AI and create a policy environment that fosters innovative private-sector solutions.

As an AI-driven healthcare technology and clinical research organization, Massive Bio specializes in decentralized clinical trials (DCTs), advanced data analytics, and the application of large language models to help sponsors, investigators, and healthcare providers rapidly match patients to clinical trials. Our **Synergy-AI** platform harnesses next-generation AI techniques for data integration, enabling sponsors to accelerate research timelines, reduce costs, and generate new discoveries more efficiently. By leveraging public-private partnerships and a modular technology architecture, we believe our work aligns closely with the Administration's priority areas for sustaining and enhancing America's AI competitiveness.

Below, we provide specific input across a range of RFI priorities—**hardware and chips**, **data centers**, **model development**, **open-source initiatives**, **explainability**, **cybersecurity**, **privacy**, **risks, regulation, and governance**, **national security**, **R&D**, **workforce development**, **innovation and competition**, **procurement**, and **international collaboration**—to remove unnecessary regulatory hurdles, support private-sector-led **innovation**, and strengthen U.S. global leadership in AI-based healthcare.

II. Massive Bio's AI-Driven Approach in Healthcare and Clinical Research

1. **Synergy-AI Platform**
    - **Comprehensive AI Ecosystem**: Our **Synergy-AI** platform uses large language models (LLMs) and advanced analytics to automate clinical trial recruitment and retention for patients in any therapeutic area.
    - **Federated Learning and Privacy**: We incorporate federated learning principles to safeguard patient data while continuously improving our algorithms. This approach meets HIPAA requirements, fosters strong partnerships, and underscores the **innovation** possible under clearly defined, pro-growth regulatory frameworks.
2. **Decentralized Clinical Trials (DCTs)**
    - **Scaling and Resilience**: Our decentralized approach leverages telehealth, mobile clinics, and remote monitoring devices to engage patients in all types of communities, improving clinical trial recruitment while reducing many of the logistical hurdles associated with traditional research.
    - **Regulatory Streamlining**: We strongly advocate for clear, flexible guidelines that encourage the **innovation** needed to optimize clinical research and deliver meaningful patient benefits without undue administrative burdens.
3. **Innovation for All Patients**
    - **Broad Patient Access**: Through partnerships with community health networks,

pharmacies, local clinics, and academic sites, Massive Bio's **hub-and-spoke** model extends trial access nationwide, advancing next-generation approaches in real-world environments.

- **Efficiency in Research**: By leveraging our platform's adaptability, we can engage patients efficiently and reliably, helping ensure no opportunity for discovery is lost due to outdated systems or restrictive policies.

III. Recommendations and Alignment with the RFI Priorities

Below, we highlight specific policy areas raised in the RFI and how Massive Bio's experiences and insights can inform the forthcoming **AI Action Plan**.

1. Hardware and Data Centers

- **Recommendation**: Strengthen **public-private collaborations** to expand access to high-performance computing (HPC) resources in healthcare. Tax incentives or matching grants would let more AI-focused firms employ advanced computing for medical research and large-scale model development.
- **Massive Bio Alignment**: Our Synergy-AI platform processes large volumes of clinical and real-world data, benefiting from HPC capacity. Upgrading HPC infrastructure **lowers computational costs** and accelerates the development of breakthrough healthcare solutions.

2. Energy Consumption and Efficiency

- **Recommendation**: Encourage **energy-efficient AI** model training and edge-computing practices that minimize data-transfer needs. Create incentives for AI developers and data center operators to adopt environmentally conscious HPC solutions.
- **Massive Bio Alignment**: By leveraging **federated learning** and on-device processing, we reduce energy use and bandwidth demands while preserving performance and data privacy.

3. Model Development and Open-Source Initiatives

- **Recommendation**: Increase federal investment in **open-source frameworks** for AI in healthcare. Establish guidelines that preserve patient privacy while motivating open data models (e.g., HL7 FHIR, OMOP) to simplify system interoperability.
- **Massive Bio Alignment**: We use open standards, such as FHIR and CDISC, to connect with existing health IT systems, thus enhancing efficiency and **innovation** in clinical trial operations.

4. Explainability, Assurance, and Trust

- **Recommendation**: Create best-practice guidelines and voluntary standards for **explainable AI** in high-stakes areas (e.g., healthcare, defense) while avoiding rigid mandates that inhibit private-sector **innovation**.
- **Massive Bio Alignment**: Our clinical approach includes real-time oversight by medical experts and curated feedback loops to maintain accuracy and ensure validated outputs from AI-driven processes.

5. Cybersecurity and Data Privacy

- **Recommendation**: Craft robust cybersecurity and privacy frameworks specific to AI in healthcare, encouraging encryption, identity management, and secure data-sharing to mitigate risks while supporting **innovation**.
- **Massive Bio Alignment**: We maintain HIPAA, GDPR, and national security compliance, using zero-trust security and robust encryption to protect patient information within distributed AI workflows.

6. Regulation and Governance

- **Recommendation**: Streamline and coordinate **federal AI guidance** (FDA, NIH, NSF, ONC) to reduce fragmentation. Clear, forward-thinking policies let companies innovate freely while meeting patient safety standards.
- **Massive Bio Alignment**: A unified, flexible regulatory ecosystem can accelerate AI-driven solutions for trial enrollment, real-world evidence, and emergency preparedness.

7. National Security and Defense

- **Recommendation**: Advance AI innovation for rapid response in potential pandemics or biosecurity events. Reinforce public-private partnerships to ensure the best emerging solutions can scale in critical situations.
- **Massive Bio Alignment**: Our capacity to deploy trials within days showcases how private-sector AI can serve national preparedness goals.

8. Research and Development

- **Recommendation**: Bolster **SBIR/STTR** programs and R&D tax incentives that specifically target AI in healthcare. These initiatives can catalyze new solutions and keep the U.S. at the forefront of biomedical **innovation**.
- **Massive Bio Alignment**: Our growth roots back to a successful NIH/NCI SBIR grant. Continued funding for these programs will foster more AI-driven healthcare breakthroughs.

9. Education and Workforce

- **Recommendation**: In partnership with academia and industry, expand training pathways in AI, data science, and interdisciplinary fields, producing a workforce ready to tackle emerging challenges.
- **Massive Bio Alignment**: Our staff encompasses clinicians, data scientists, developers, and operational specialists. We rely on a steady pipeline of skilled professionals trained in AI for healthcare applications.

10. Innovation and Competition

- **Recommendation**: Protect U.S. intellectual property while enabling open collaboration. Federal policy must balance the need for strong IP protections with the benefits of an **innovation**-driven collaborative ecosystem.
- **Massive Bio Alignment**: We work with various organizations under flexible IP arrangements that expedite the practical application of AI solutions, helping maintain a competitive U.S. leadership role.

11. Federal Procurement

- **Recommendation**: Reform rules to let federal health agencies quickly adopt private-sector AI tools. Develop transparent scoring methods to measure efficacy, cost-savings, and compliance without locking out smaller innovators.
- **Massive Bio Alignment**: We have worked under multiple contract vehicles. Streamlining these processes enables swift deployment of transformative AI in large-scale public programs.

12. International Collaboration and Export Controls

- **Recommendation**: Protect advanced U.S. AI technologies from hostile actors through carefully calibrated export controls, while promoting responsible global AI collaborations to drive medical **innovation**.
- **Massive Bio Alignment**: Massive Bio operates in 17 countries but ensures rigorous data safeguards. Clear guidelines for international AI engagements will sustain global trust in U.S.-developed innovations.

IV. Conclusion and Path Forward

Massive Bio commends the Administration's focus on **removing regulatory barriers** and supporting a strong private-sector environment that cultivates **innovative AI solutions**. Our deep experience in clinical trial operations, decentralized research, large language model applications, and advanced partnerships demonstrates how policy frameworks prioritizing innovation can accelerate healthcare breakthroughs.

We encourage policymakers to **strengthen public-private partnerships**, **invest in AI infrastructure**, and **streamline regulations** so that modern AI can thrive. By promoting R&D incentives, HPC access, and consistent national guidelines, the United States can firmly uphold its global leadership position in AI, especially in healthcare technology, clinical research, and beyond.

Thank you for considering our perspectives on the **AI Action Plan**. We look forward to collaborating with the NSF, the White House, and other stakeholders to ensure the United States remains at the forefront of AI **innovation** and scientific discovery.

Submitted on behalf of:
Massive Bio, Inc.
90 West Street #12M, New York, NY 10006, USA
Point of Contact:

- **Name**: Dr. Arturo Loaiza-Bonilla, MD, MSEd, FACP (Co-Founder & Chief Medical Officer)

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